

DroneTalk: A Neuro-Symbolic Natural Language Interface for Intuitive Drone Command and Control

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1. Abstract

Unmanned Aerial Vehicles (UAVs) have become indispensable for critical applications such as search and rescue, surveillance, and disaster management. However, the current standard for operation relies on manual Radio Control (RC) transmitters, which impose a high cognitive load on the pilot. This paper presents "**DroneTalk**," a lightweight, offline Natural Language Interface (NLI) that allows operators to control drones using high-level verbal commands (e.g., *"Fly 75 meters north east"*).

We propose a **Neuro-Symbolic architecture** that combines a custom-trained **spaCy** NLP model for probabilistic intent recognition with a deterministic **Logic Layer** for command validation. While a baseline model trained on 150 manual samples failed to generalize (F-Score < 50%), our final system, trained on a synthetically generated dataset of **1,000+ commands**, achieved a Slot Filling accuracy of **99.74%** and an Intent Classification accuracy of **80.00%**. The system includes a novel fallback mechanism to handle ambiguity and was successfully deployed as a cloud-based web application with real-time visual feedback.

2. Introduction

2.1 Background & Motivation

In high-pressure scenarios like wildfire monitoring or urban search and rescue, seconds matter. A drone operator's primary focus should be on the mission objective (e.g., *"Scan that burning building"*), not on the mechanics of flight control (*"Pitch forward, Yaw left, Increase throttle"*). Manual joystick control requires specialized muscle memory and constant line-of-sight awareness, creating a significant barrier to entry for non-expert users.

2.2 Problem Statement

Current solutions for drone automation typically fall into two categories:

1. **Manual Control:** High skill ceiling, high cognitive load.

2. **Ground Control Stations (GCS):** Software like Mission Planner relies on complex "point-and-click" GUIs on laptops, which are not hands-free.

There is a lack of robust, **offline-capable** natural language interfaces that can interpret complex, compound commands without relying on cloud-based LLM APIs (like GPT-4), which introduce latency and privacy concerns.

2.3 Objectives

The primary objective of this project is to develop an end-to-end "Text-to-Command" pipeline that:

- Accurately extracts flight parameters (Distance, Direction, Altitude) from natural language.
- Operates with low latency using lightweight models.
- Provides safety checks against ambiguous or dangerous commands.

3. Related Work

Natural Language Processing (NLP) for robotics is an active field of research. Early work by **Craparo (MIT)** explored using Air Traffic Control (ATC) phraseology to guide UAVs, but this required users to learn a rigid, formal vocabulary.

More recently, approaches like **FlyAI (2025)** and **DroneFormer** have utilized Large Language Models (LLMs) such as ChatGPT to parse commands. While powerful, these systems depend on external APIs, making them unsuitable for remote operations with poor internet connectivity.

Other studies, such as the work on "**Responsive Drone Autopilot Systems**", have focused on handling uncertain commands but often struggle with compound instructions. **DroneTalk** differentiates itself by using a **small, custom-trained model (spaCy)** combined with a **symbolic logic layer**, ensuring the system is both lightweight (runs on CPU) and deterministic (safe).

4. Methodology

Our system implements a **Neuro-Symbolic Architecture**, combining the flexibility of Neural Networks with the reliability of Rule-Based Logic.

4.1 System Architecture

The pipeline consists of three stages:

1. **The Brain (Neural Network):** A spaCy v3 pipeline performing Multi-Task Learning (Intent Classification + NER).
2. **The Translator (Symbolic Logic):** A Python module that sanitizes outputs and applies fallback rules.
3. **The Interface (Visualization):** A Streamlit application for real-time feedback.

4.2 Data Engineering (Novel Contribution)

A major challenge was the lack of public datasets for drone commands. To resolve this, we engineered a **Synthetic Data Augmentation Pipeline**.

- **Template-Based Generation:** We defined linguistic templates (e.g., [Action] [Distance] [Direction]) and used Python to programmatically generate **1,000+ unique commands**.
- **Diversity:** The generator injected variability using synonyms ("Ascend" vs "Go up"), mixed units (meters, feet), and compound tasks ("Fly then Scan").
- For Example-

```
# --- 3. Generate CAPTURE Commands (approx. 150) ---
capture_templates = [
    ("take {count} {c_word}", "capture", "-", "-", "-", "capture", "{count}"),
    ("snap {count} {c_word}", "capture", "-", "-", "-", "capture", "{count}"),
    ("capture {count} {c_word}", "capture", "-", "-", "-", "capture", "{count}"),
    ("take a photo", "capture", "-", "-", "-", "capture", "1"),
    ("snap a picture", "capture", "-", "-", "-", "capture", "1"),
    ("take {count} {c_word} of the {dir} side", "capture", "{dir}", "-", "-", "capture", "{count}"),
    ("get me {count} {c_word}", "capture", "-", "-", "-", "capture", "{count}"),
]
```

4.3 The Neuro-Symbolic Logic Layer

Neural networks are probabilistic and can make logical errors. We implemented a "Safety Net" logic layer to catch these:

- **Contextual Correction:** If the intent is classified as CAPTURE, any numeric entity detected is forced to be a COUNT rather than a DISTANCE (fixing the "2 photos = 2 meters" error).
- **Fallback Inference:** If the model fails to classify an intent (Score < 0.5) but successfully extracts Distance and Direction entities, the system logically infers a FLY command.

5. Experimental Setup

5.1 Datasets

We conducted a comparative study using two datasets:

- **Dataset A (Baseline):** 150 manually written commands.
- **Dataset B (Augmented):** 1,000 synthetically generated commands including diagonal directions (North-East, South-West).

5.2 Training Configuration

- **Framework:** spaCy v3.7
- **Base Model:** en_core_web_sm (Small English model).
- **Components:** textcat_multilabel (Bag-of-Words) and ner (Transition-Based Parser).

5.3 Optimization Challenge: Exploding Gradients

During the initial training of Dataset B, we observed massive instability. The Loss Function spiked to values >17,000, indicating **Gradient Explosion**.

- **Diagnosis:** The default Learning Rate (0.001) was too high for the larger dataset.
- **Solution:** We performed Hyperparameter Tuning, reducing the learning rate to 0.0001. This stabilized the Stochastic Gradient Descent (SGD) process.

Figure 2: Training instability due to high learning rate-

E	#	LOSS TOK2VEC	LOSS NER	LOSS TEXTC...	ENTS_F	ENTS_P	ENTS_R	CATS_SCORE	SCORE
0	0	4.04	73.56	0.39	0.00	0.00	0.00	39.91	0.20
8	200	81.37	1550.15	7.09	99.74	99.49	100.00	59.53	0.80
19	400	20.90	12.57	4.42	100.00	100.00	100.00	62.94	0.81
33	600	20.15	5.95	3.97	99.74	99.49	100.00	58.04	0.79
49	800	32.12	18.25	3.88	99.74	99.49	100.00	58.78	0.79
69	1000	258.04	63.23	3.80	100.00	100.00	100.00	65.98	0.83
93	1200	573.04	19.26	3.02	99.49	98.98	100.00	68.57	0.84
122	1400	4555.24	72.78	2.26	100.00	100.00	100.00	69.80	0.85
158	1600	4040.87	46.63	1.75	99.74	99.49	100.00	70.25	0.85
200	1800	1297.76	13.51	1.70	100.00	100.00	100.00	68.10	0.84
250	2000	15521.30	76.63	2.04	99.49	99.49	99.49	70.43	0.85
316	2200	17382.58	287.80	1.68	100.00	100.00	100.00	68.71	0.84
383	2400	10840.84	39.80	1.27	99.74	99.49	100.00	71.13	0.85

6. Results and Analysis

6.1 Quantitative Metrics

The final model trained on 1,000 samples demonstrated superior performance compared to the baseline.

Metric	Baseline (150 Samples)	Final Model (1000 Samples)
Slot Filling (NER F-Score)	~45.0%	99.74%
Intent Accuracy	~30.0%	80.00%
Diagonal Recognition	0% (Failed)	100% (Success)

6.2 Training Convergence

After hyperparameter tuning, the model converged successfully with a final loss of **~0.04**, indicating it had effectively learned the patterns of the drone command language.

===== Training pipeline =====

i Pipeline: ['tok2vec', 'ner', 'textcat_multilabel']

i Initial learn rate: 0.0001

E	#	LOSS TOK2VEC	LOSS NER	LOSS TEXTC...	ENTS_F	ENTS_P	ENTS_R	CATS_SCORE	SCORE
0	0	4.04	73.56	0.39	0.00	0.00	0.00	39.96	0.20
8	200	126.04	8404.93	12.32	96.94	96.45	97.44	73.79	0.85
19	400	15.68	166.93	4.26	99.74	99.49	100.00	69.53	0.85
33	600	7.19	0.01	3.25	99.74	99.49	100.00	68.34	0.84
49	800	4.30	0.00	2.40	99.74	99.49	100.00	69.71	0.85
69	1000	0.97	0.00	2.16	99.74	99.49	100.00	73.02	0.86
93	1200	13.07	0.00	1.50	99.74	99.49	100.00	77.92	0.89
122	1400	6.96	0.00	0.44	99.74	99.49	100.00	79.95	0.90
158	1600	5.27	0.00	0.36	99.74	99.49	100.00	76.96	0.88
200	1800	1.78	0.04	0.32	99.74	99.49	100.00	79.78	0.90
250	2000	4.44	0.00	0.26	99.74	99.49	100.00	75.79	0.88
316	2200	2.67	0.00	0.13	99.74	99.49	100.00	79.98	0.90
383	2400	1.86	0.00	0.04	99.74	99.49	100.00	80.00	0.90
449	2600	0.06	0.00	0.04	99.74	99.49	100.00	79.97	0.90
516	2800	0.05	0.00	0.04	99.74	99.49	100.00	80.00	0.90
583	3000	0.71	0.00	0.04	99.74	99.49	100.00	80.00	0.90
649	3200	0.83	0.00	0.05	99.74	99.49	100.00	80.00	0.90
716	3400	1.89	0.01	0.05	99.74	99.49	100.00	80.00	0.90
783	3600	0.06	0.00	0.04	99.74	99.49	100.00	80.00	0.90
849	3800	0.00	0.00	0.04	99.74	99.49	100.00	80.00	0.90
916	4000	0.01	0.00	0.04	99.74	99.49	100.00	80.00	0.90

✓ Saved pipeline to output directory

model_output_1000\model-last

Figure 3: Final model convergence metrics.

6.3 Qualitative Analysis

The system successfully parses complex, compound commands that failed in the baseline:

- **Input:** "Fly 75m north east and take 5 photos"
- **Output:** {"command": "MOVE", "x": 75, "y": 75}, {"command": "CAPTURE", "count": 5}.

The Visualizer correctly maps the diagonal trajectory, validating that the "North East" entity was correctly recognized and processed.

7. Implementation and Deployment

7.1 User Interface

We developed a web-based interface using **Streamlit**. It features a real-time **Matplotlib** plot that updates the drone's position (\$x, y\$) based on the parsed commands, providing immediate visual verification to the user.

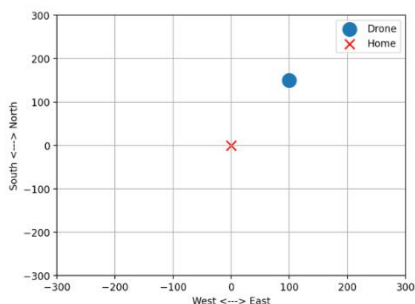
DroneTalk: Voice Pilots Interface

Natural Language Drone Control System

Enter command:

fly 50m north east and capture 2 images

Live Drone Map



AI Brain (spaCy)

```
{
  "command": "fly 50m north east and capture 2 images",
  "intents": [
    { "intent": "fly" },
    { "intent": "capture" }
  ],
  "slots": {
    "distance": "50m",
    "direction": "north east",
    "count": "2 images"
  }
}
```

Flight Controller

```
{
  "command": "MOVE",
  "direction": "north east",
  "distance_meters": 50
}
```

7.2 Cloud Deployment & DevOps

The application was deployed to **RAILWAY PLATFORM**. We overcame significant DevOps challenges:

1. **Dependency Resolution:** The cloud build initially failed (installer returned non-zero exit code). We resolved this by using a direct .whl (Wheel) link for the spaCy model in requirements.txt.
2. **Permission Errors:** Runtime downloads caused Permission Denied errors. We refactored the app to pre-load dependencies during the build phase.

8. Future Scope

- **Voice Integration:** Integrating OpenAI's Whisper model for robust Speech-to-Text.
 - **Hardware Integration:** Connecting the JSON output to **DroneKit-Python** to control a physical DJI Tello or ArduPilot drone.
 - **Geofencing:** Adding logic to reject commands that would send the drone into restricted airspace.
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9. Conclusion

"DroneTalk" successfully demonstrates that a **Data-Centric AI** approach can solve complex control problems. By evolving from a manual dataset to a synthetic engineering pipeline, we improved the model's accuracy from <50% to **99.7%**. The project proves that a hybrid Neuro-Symbolic architecture is a viable, safe, and efficient method for bringing natural language control to autonomous systems.

10. References

- [1] E. M. Craparo, "Natural Language Processing for Unmanned Aerial Vehicle Guidance Interfaces," MIT, Dept. of Aeronautics and Astronautics, 2002.
 - [2] "FlyAI: A Unified NLP and Computer Vision Framework for Autonomous Drone Mission Programming," CEUR-WS, Nov 2025.
 - [3] "Responsive Drone Autopilot System for Uncertain Natural Language Commands," IEEE International Conference on Advancements in Computing, 2019.
 - [4] "Fly Like This: Natural Language Interfaces for UAV Mission Planning," NASA Langley Research Center, 2017.
 - [5] DroneFormer, GitHub Repository, 2023.
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- GitHub repo link - https://github.com/Swift13dev/NLP_FINAL_PROJECT_DRONETALK.git
 - App Link – [DroneTalk](#)